Handheld & Embedded Calculator

Production-ready handheld electronics - custom firmware, PCB design, and a 3D-printed enclosure.

ADRIANO ZAGAR

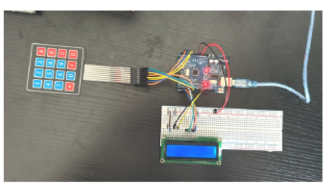
C++ / ARDUINO / ESP32 / KICAD / SOLIDWORKS

FIRMWARE	Custom C++ computational logic
CONTROL	ESP32 microcontroller
DISPLAY	4-pin LCD interface
HARDWARE	2-layer PCB / MT3608 / TP4056 / 3.7 V rechargeable battery
ENCLOSURE	SOLIDWORKS snap-fit housing
GOAL	Compact, cost-efficient product

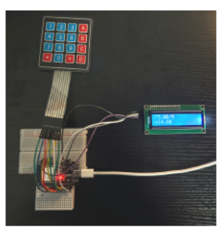
Final assembled handheld calculator

BUILD PROCESS

01	02	03	04	05	06
PROTOTYPE	**MIGRATE**	**VALIDATE**	**SCHEMATIC**	**PCB**	**ENCLOSURE**
Arduino Uno	ESP32 system	Perf-board	KiCad	2-layer board	SOLIDWORKS

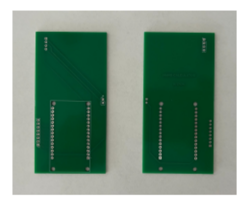


Arduino Uno Breadboard



ESP32 Breadboard

Perf-Board



Custom PCB

ENGINEERING HIGHLIGHTS

Cross-domain integration

Firmware, electronics, and enclosure

Spatial optimization

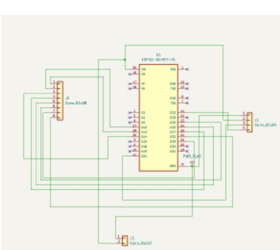
Compact handheld architecture

Design for manufacturing

Efficient assembly and sourcing

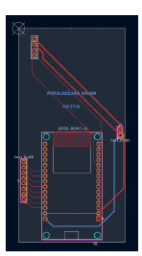
Rechargeable power system

MT3608 / 3.7 V battery / TP4056



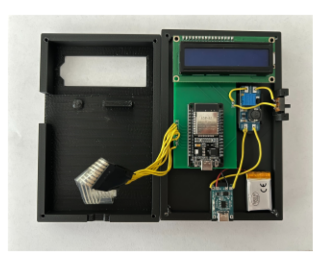
Schematic

KiCad



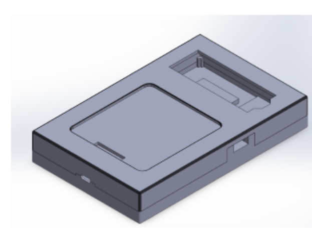
PCB Layout

KiCad



Internal Electronics

Rechargeable power system



Enclosure CAD

SOLIDWORKS